

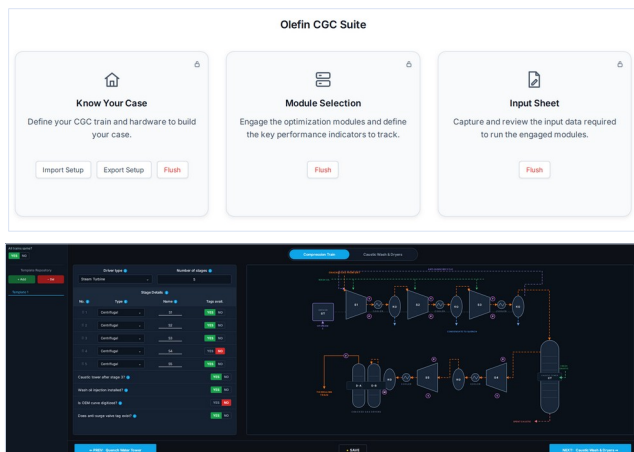
## PRODUCT OVERVIEW

CGCX360 is a decision-support application for the cracked gas compressor (CGC) train. Domain and ML agents are configured on the unit's own historian and sensor tag data, covering stage-wise polytropic efficiency and head, surge / anti-surge margin, interstage cooler and knock-out drum performance, wash oil / water injection for fouling control, steam turbine drive efficiency, caustic / amine wash tower and cracked gas dryers. Output reaches the process engineer as ranked, quantified actions, each reported against the parameter that caused it, balancing compression power against furnace throughput. Every model agent is configured, validated and commissioned per site by Ingenero process engineers, so the deployment reflects that unit's configuration and operating envelope.



### 01 Unit Configuration

Shown: Olefin CGC suite - interactive configuration, modules and sensor entry

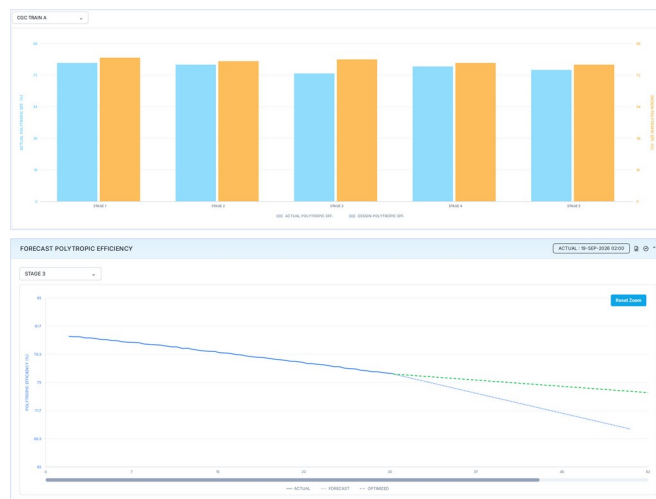


- Know Your Case** Configure stages, coolers, KO drums, driver, caustic tower and dryers
- Module Selection** After configuring the CGC train, select the modules - Domain & AI agents
- Input Sheet** Based on configuration & Module selection - relevant sensor info needs to be filled
- Interactive Config** Compressor stages, turbine, caustic wash & dryer interactive PFD available



### 02 Plant Data and Model Execution

Shown: stage-wise polytropic efficiency, efficiency forecast current vs optimum.



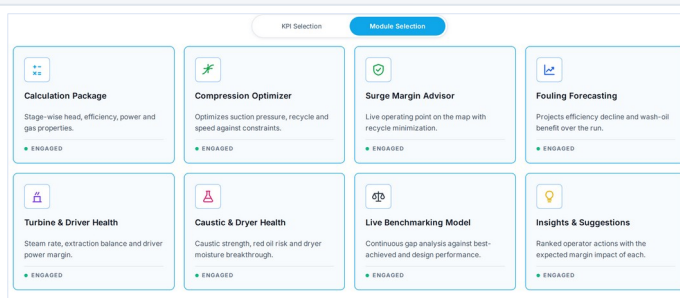
- Inputs** Sensor tag entry with design / OEM performance curves and configuration
- Processing** tag validation, steady state framing and gas mol. weight estimation
- Outputs** predicted KPIs, ranked actionable with cause, trends, alerts



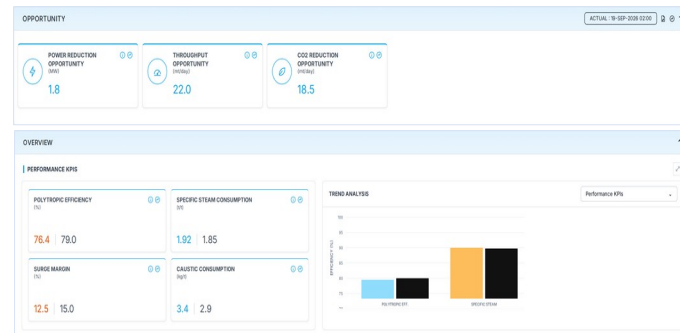
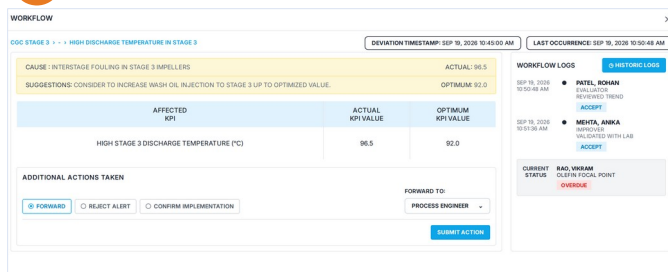
### 03 AI-Powered Model Agent Suite

Agent/Module selection shared along with detailing of the CGC specific details

- Compressor Performance** Stage-wise polytropic efficiency and head against OEM curves.
- Surge Margin Monitoring** Tracks operating point vs surge line and minimizes recycle.
- Interstage Fouling** Detects fouling and optimizes wash oil / wash water injection.
- Turbine Drive Efficiency** Estimates steam rate, extraction and driver power margin.
- Caustic Wash Optimization** Optimizes caustic strength and flags CO<sub>2</sub>/H<sub>2</sub>S slip and red oil.
- Dryer Cycle Optimization** Predicts moisture breakthrough and optimizes regeneration timing.
- Machine Health (ML)** Anomaly detection on vibration, bearing and seal-gas trends



### 04 Opportunity and Operator Actions



2-4%

**Compression power**  
polytropic efficiency & recycle optimization

+1-3%

**CGC throughput**  
surge & suction pressure optimization

10-15%

**Caustic consumption**  
spent caustic strength optimization

Basis: optimization respects compressor, furnace throughput and cold section limits; suggestions take care of step change for smooth implementation.



### 05 Platform

- Historian connectivity** plug and play with leading plant historians
- Simulation integration** third party simulators integrate without custom work
- Daily reporting** downloadable daily reports with interactive trends
- Gen AI assistant** answers plant questions in the operating workflow
- Build your own dashboard** engineers build the views their unit needs

#### DELIVERY

Configured, validated and commissioned by Ingenero process engineers.

#### FLEXIBILITY

4 or 5 stage trains; steam or motor drive; selectable KPI set

#### SCALE

Repeat sites reuse the configuration, with no/minimal rework per plant.

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